WEKA Classification Assignment 01

Name: Abdiashim Amirov **Group:** MAN-2 **Student ID:** _____

Task 1: Dataset exploration

The iris dataset has **150 instances** and **5 attributes**: four numeric predictors (sepallength, sepalwidth, petallength, petalwidth) and the nominal class. The class has three labels, Iris-setosa, Iris-versicolor and Iris-virginica, with **50 instances each**, so the data is perfectly balanced. There are **no missing values** (0%). In the visualization, Iris-setosa is clearly separated from the other two by small petal length and width, while versicolor and virginica form a rising band that partly overlaps. Petal measurements therefore look most useful for classification.

Task 2: Training and evaluation

Both classifiers (J48 and NaiveBayes) were run in WEKA 3.8.7 with default settings, no filters, and 10-fold cross-validation (random seed 1), class attribute = class.

Model	Accuracy (%)	Weighted average F1
J48	96.00 (144/150)	0.960
NaiveBayes	96.00 (144/150)	0.960

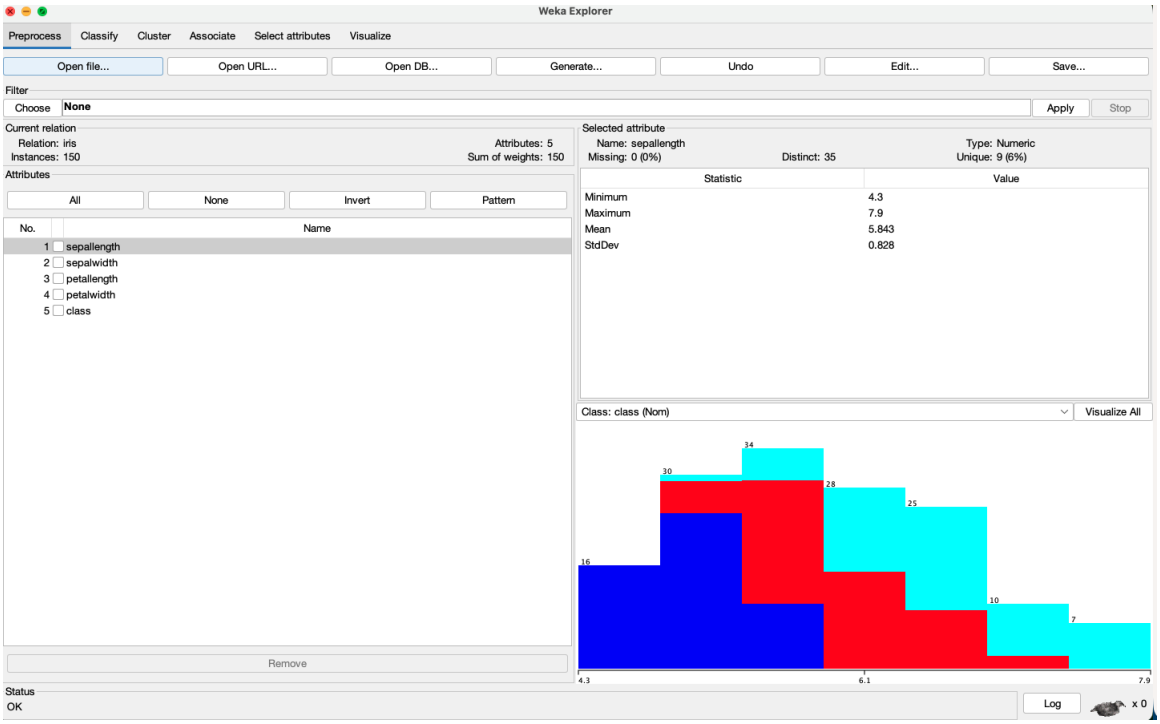
Task 3: Interpretation

J48 confusion matrix (rows = actual): setosa 49/1/0, versicolor 0/47/3, virginica 0/2/48. **NaiveBayes:** setosa 50/0/0, versicolor 0/48/2, virginica 0/4/46. For both models the most confused pair is **Iris-versicolor and Iris-virginica** (J48: 3+2 errors; NaiveBayes: 2+4), which matches the overlap seen in the plots. J48 also misclassified one setosa as versicolor. One J48 rule: *petalwidth <= 0.6 then Iris-setosa* (50 instances, no errors). The tree shown in the output is built on the full dataset and is separate from the ten models trained during cross-validation; the reported accuracy comes from the cross-validation.

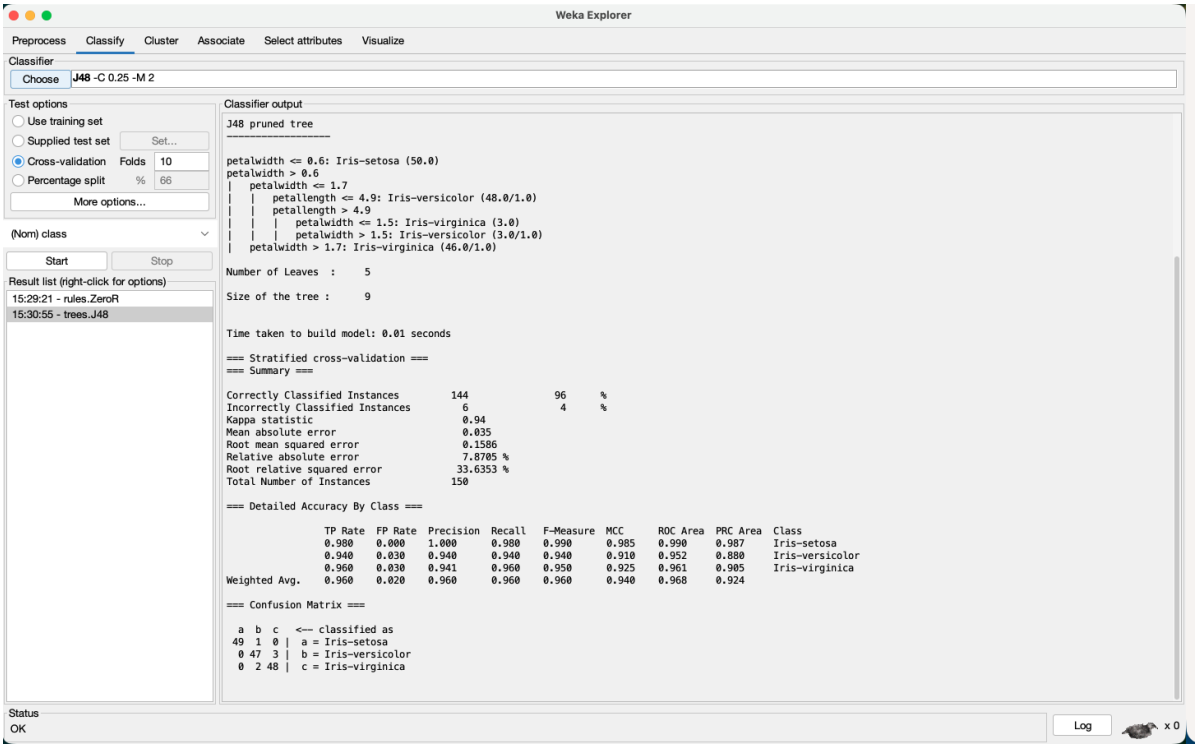
Task 4: Recommendation

Both models achieve identical accuracy (96%) and weighted F1 (0.960) under 10-fold cross-validation, so these headline metrics cannot separate them. I would recommend **J48** for this dataset. Its main advantage is interpretability: it produces a small tree of 5 leaves that can be read as rules, for example petalwidth <= 0.6 identifies setosa, and petal length and width decide between versicolor and virginica. This is easy to explain to non-technical users and to audit. NaiveBayes has some merits: it classified setosa perfectly, had slightly lower RMSE (0.155 vs 0.1586) and a higher weighted ROC area (0.994 vs 0.968), meaning better probability estimates. However, it assumes independent attributes, although petal length and width are strongly correlated, and its outputs are less transparent. Both train in about 0.01 seconds, so speed does not matter here. Since predictive performance is tied, transparency decides in favour of J48, although with only 150 instances a larger test would be needed to confirm any difference.

Screenshot 1: Dataset summary (Preprocess tab)



Screenshot 2: J48 evaluation output



Screenshot 3: NaiveBayes evaluation output

